TIL6022 Project Report: Evaluating External Context Features in Short-Term Crowd Flow Forecasting for SAIL Amsterdam

Project Group 2
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1. Introduction

Events involving large crowds will bring people together in a relatively limited space, leading to rapidly changing pedestrian flow and congestion problems. SAIL Amsterdam is an example of such an event, with large ships coming from all over the world in and around the IJ harbor of Amsterdam. The 2025 edition saw very large crowds of visitors to the event location, making it necessary to implement crowd management measures for keeping things safe and accessible. The use of short-term crowd predictions may help in this respect, as they give information on expected crowd conditions before congestion becomes an issue.

For SAIL 2025, pedestrian flows have been measured via sensors positioned along key walking paths, giving data on pedestrian numbers that pass by the sensors in short time intervals and in both directions. Hence, there is information available on how pedestrian flows evolve over time and in certain locations. Additionally, abrupt changes in pedestrian flows may occur as a consequence of context factors outside the scope of the crowd data. Next to crowd sensor data, other data is available which could potentially be of relevance when looking at short-term crowd predictions, including meteorological data \cite{daamen2026sail}.

This report focuses on exploring the effectiveness of using external contextual information to enhance short-term crowd forecasting. More precisely, this work aims at addressing the following research question: "To what extent does the integration of external context factors improve the accuracy of short-term crowd predictions for large-scale public events such as SAIL?" The importance of factors such as weather and event program highlights will be explored and tested against a baseline approach based on crowd observations from the past \cite{li2026reliable,hari2026crowd}.

2. Research Objective & Research Questions

The objective of this research is to investigate whether the integration of external contextual factors improves the accuracy and practical applicability of short-term pedestrian-flow forecasts during the

large-scale public event SAIL 2025. An auto regressive baseline model will be developed using historical sensor observations. This model will be compared with a context-enriched model that incorporates information such as weather conditions and scheduled SAIL activities.

Both models will be evaluated at selected sensor locations for forecast horizons of 15, 30 and 45 minutes. Their forecast accuracy will be evaluated using the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The individual added value of each category of contextual information will be investigated by systematically adding and removing groups of contextual variables. Finally, the operational usefulness of the context-enriched model will be evaluated by examining its ability to provide timely warnings of upcoming periods of high pedestrian flow while minimising the number of false alarms.

In order to address the research objective, the following research questions were formulated. Using SAIL 2025 as a case study, these questions focus on the contribution of external contextual factors to short-term pedestrian-flow forecasting, the accuracy of the prediction models, and their performance under different conditions.

Main Research Question

"To what extent does the integration of external context factors improve the accuracy of short-term crowd predictions for large-scale public events such as SAIL?"

Sub-questions

1. Feature Importance & Ranking

Which external context factors (weather conditions and vessel activity) contribute most significantly to the predictive performance of the model?

This subquestion focuses on gaining an insight into which context factors play a role in pedestrian flows. By analysing the importance of weather conditions and vessel activity, the study aims to identify which factors provide the most useful additional information.

2. Added Value over Baseline across Prediction Horizons

How does the predictive accuracy (measured in RMSE and MAE) of a context-enriched model compare to a purely autoregressive baseline across prediction horizons of 15, 30 and 45 minutes?

The second subquestion investigates if incorporating external context information plays a role in the accurateness and how much compared to using the historical flow data alone. The comparison is made over different time horizons in order to investigate whether there is any added value over changes of time.

3. Robustness & Generalizability (Peak vs. Regular Days)

To what extent does the forecasting model maintain its accuracy when evaluated on unseen event days with extreme conditions (e.g., the Sail-In parade or acute weather changes)?

The last subquestion focuses on exploring how well the forecasting models perform under the different conditions during the SAIL 2025 event.

3. Data Used

The data used for this project was obtained from the faculty, called 'SAIL2025_LVMA_data_3min_20August-25August2025_flow.csv', 'sensor-location.xlsx' and ‘SAIL_Amsterdam_Weather_Data.csv’ and ‘Vesselposition_data_20-24Aug2025.csv’

All figures and graphs that are used in the report, are created based on the provided data.

The dataset exhibits a clear daily cycle, with sensors showing similar behaviour and peaks occurring at roughly the same times. Therefore, we combined data from multiple days to improve prediction accuracy. Since the data per sensor was also behaving similarly, we filtered the data to only 3 sensors instead of 36 to give a clearer and better outline.

Sensor location data
Geographic coordinates and metadata for all crowd sensors deployed during SAIL 2025. This reference data enables spatial analysis and visualization of crowd patterns. The chosen sensor locations are visible in figure 1.

Figure 1: sensor locations



Crowd sensor data
Real-time crowd flow measurements from multiple sensors placed throughout the Amsterdam event area during SAIL 2025. The sensors measure pedestrian flow in different directions, providing bidirectional crowd movement data.

Weather data

The weather observations were received from the Royal Netherlands Meteorological Institute (KNMI) for Amsterdam during the SAIL 2025 event. These were used to examine potential influence on pedestrian flows during the event. The datasets consist of variables like temperature, relative humidity and precipitation. All of these may influence the flows during the event. Therefore, they are useful to analyse.

The weather data used for the project was on the Friday of the event, from 16:00-18:00 hours. This day was chosen on the basis that the data provided was useful data which could be analysed and used for the model.

Vessel position data
Vessel position data is used in this project to investigate the relationship between vessel activity and pedestrian flows during the event. Special ships, such as tall ships, may attract visitors and influence pedestrian flows at nearby locations. The used dataset contains information on vessel locations and movements. This allows for analysis on whether the presence of certain ships near pedestrian sensors is associated with changes in pedestrian flows. This provides insight into whether vessel activity could be a useful factor for improving short-term crowd forecasts.

4. Data Pipeline

The data pipeline starts with collecting and preparing the different datasets used in the analysis. The vessel position dataset was cleaned before further processing by removing vessel types that were not relevant to the study, such as ferries and other non-event-related traffic. This reduced noise in the dataset and made it more suitable for analysing vessel movements related to SAIL 2025. After cleaning, the vessel data can be combined with the crowd sensor, weather and event programme data based on location and time, allowing the different data sources to be analysed together.

Vessel methodology

We link the position of 20 tall ships and sailing vessels to three crowd sensors, to test whether vessel activity improves short-term forecasts of pedestrian flow.

The vessel file covers Friday August 22 2025, 16:00-18:00 local time. We selected only the SAIL fleet, the ships most likely to attract visitors and only for this period; ferries and other non-event traffic are not included. It holds 1567 position reports (pings) for 20 ships, each with a timestamp, name, category, length (m), speed (m/s), longitude and latitude. Sixteen ships are classed as sail training ships and four as sailing vessels. Lengths run from 23-115 m and six ships are 60 m or longer. Ships report irregularly: the median gap between two pings of one ship is 90 s, and the longest gap is 54 minutes. Using the project's threshold of 0.5 knot, 33% of the pings are from moving ships; most ships lay still. Only WYLDE SWAN, TWISTER and AVONTUUR were moving in most of their pings.

Sensor coordinates and effective widths come from the base dataset: GASA-03 (8 m), GASA-04 (7 m) and CMSA-GAWW-17 (4.7 m). The sensors are 513 m
(GASA-03 to GASA-04), 663 m (GASA-04 to CMSA-GAWW-17) and 1,113 m (GASA-03 to CMSA-GAWW-17) apart.

Preprocessing

1. Timestamps are converted to UTC, so that vessels, sensors and weather share one clock.
2. Pings without position are dropped.
3. Pings are floored to the 3-minute grid of the crowd sensors. Within a bin, the positions of one ship are averaged. The window gives 40 bins per sensor, 120 sensor-time rows in total.
4. The vessel features are joined to the crowd counts (flow and flow_per_m) on sensor and timestamp.

For every ping, and for every 3-minute bin, we compute the great circle (haversine) distance in metres from the ship to each of the three sensors, with an Earth radius of 6,371,000 m:

$$d = 2R\arcsin\sqrt{\sin^2(\tfrac{\Delta\varphi}{2}) + \cos\varphi_1\cos\varphi_2\sin^2(\tfrac{\Delta\lambda}{2})}$$ (*Haversine - Distance*, n.d.)

where ϕ_1, ϕ_2 denote the sensor and vessel latitudes in radians, $\Delta\phi$ and $\Delta\lambda$ represent the differences in latitude and longitude, and r = 6371 km is the Earth's mean radius. Using this metric, we computed the instantaneous number of active vessels within critical spatial buffers (250m, 500m, and 1000m) around each crowd counting post, alongside the Euclidean/Haversine distance to the nearest moving vessel.

In the 3-minute features, distances are capped at 1,000 m, assuming that ships further away have no influence on the crowd flow at a sensor.
The distance is a continuous variable and the counts can be turned into a binary variable (a ship within the radius or not).

5. Results

5.1 Ridge Regression visualisation

Data preparation

The Ridge Regression analysis was conducted for sensor GASA-03 (Oosterdoksbrug). The used dataset spans from 20 to 24 August 2025 and contains pedestrian-flow observations at three-minute intervals. Flow measurements were aggregated into non-overlapping 15-minute counts by summing five valid observations per interval. Incomplete intervals were excluded. Temperature, humidity, and rainfall measurements were aligned to a 15-minute grid using the available observations and time-based interpolation only between observed values.

A limitation is that meteorological measurements cease on 22 August at 18:39, while the test period falls on 23–24 August. The weather-enriched model therefore cannot be validly tested on the specified holdout period without additional weather observations. The out-of-sample results described in this chapter refer to Ridge Regression without weather features.

Feature engineering: lags, rolling dynamics, and targets

Historical flow was represented by autoregressive lags at 15, 30, 45, and 60 minutes. A one-hour rolling mean and rolling standard deviation captured recent flow level and variability. Calendar features included hour, weekday, and a weekend indicator. The predictive targets represented pedestrian flow 15, 30, and 45 minutes ahead, respectively. The full specification also included temperature, humidity, and precipitation when observations were available.

Train-test split

The training period extended from 20 August 2025, 01:00, to before 23 August 2025, 23:30; the test period ran from 23 August 2025, 23:30, to 24 August 2025, 23:00. Observations without sufficient lag history or a complete 15-minute flow count were excluded. Training samples were only retained when their forecast target preceded the test-period boundary. As a result, the effective training sample sizes were 344, 343, and 342 for 15-, 30-, and 45-minute horizons, while the respective test sample sizes were 94, 93, and 92.

Model training and performance comparison across horizons

Ridge Regression with standardized input variables and fixed regularization alpha = 10 was trained separately for each forecast horizon. A naive persistence forecast, which assumes that the current 15-minute flow continues at the future horizon, was included as a reference.

Predictions were evaluated using mean absolute error (MAE) and root mean squared error (RMSE), both expressed as pedestrians per 15-minute interval.

Forecast horizon	Ridge MAE	Ridge RMSE	Persistence MAE	Persistence RMSE
t + 15 min	184.33	275.27	111.52	181.44
t + 30 min	231.75	323.51	164.67	264.47
t + 45 min	282.98	373.86	211.38	334.90

Table 5.1: Ridge Regression values

Performance analysis

Both MAE and RMSE increased with the prediction horizon, indicating reduced accuracy for longer-term predictions. Ridge Regression produced an MAE of 184.33 at the 15-minute horizon, rising to 282.98 at the 45-minute horizon. For this test period, the naive persistence baseline outperformed the Ridge model at all horizons. Consequently, the current results do not demonstrate an advantage of the specified Ridge model over the simpler benchmark. Because weather observations are unavailable for the test period, a quantitative claim about the predictive value of weather cannot be made from the dataset.

Feature importance analysis

A Ridge Regression does not yield gain-based feature importance. Instead, the absolute values of standardized Ridge coefficients were normalized to provide a descriptive ranking. For the 15-minute weather-enriched Ridge model trained on the subset with observed weather, the largest relative coefficient magnitudes were the one-hour rolling mean being 24.7%, the 15-minute flow lag (23.3%), and the 30-minute flow lag (21.2%). Temperature was ranked next (8.8%). These are associations within the fitted training model, not causal effects or evidence that weather improved test-set accuracy. Coefficient-based rankings may also be sensitive to correlations between predictors.

Interpretation limits

This analysis does not reproduce the numerical Ridge/XGBoost comparison in the existing report. Exact comparison requires both algorithms to be rerun on identical data, feature sets, preprocessing, and test observations. Most importantly, weather data covering the test period must be supplied before a weather-enriched model can be evaluated on those dates.

Figure 5.1: Ridge Regression with weather

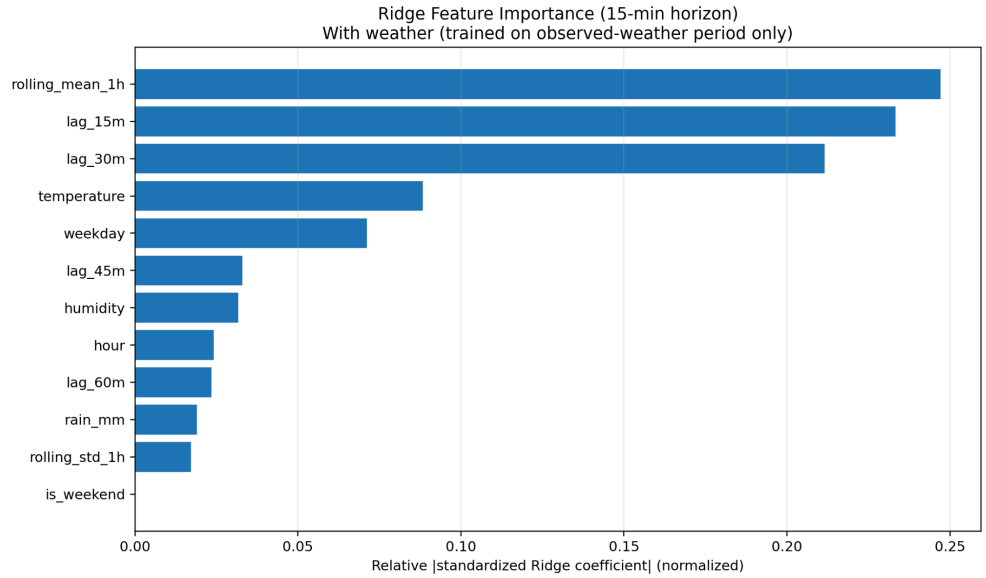
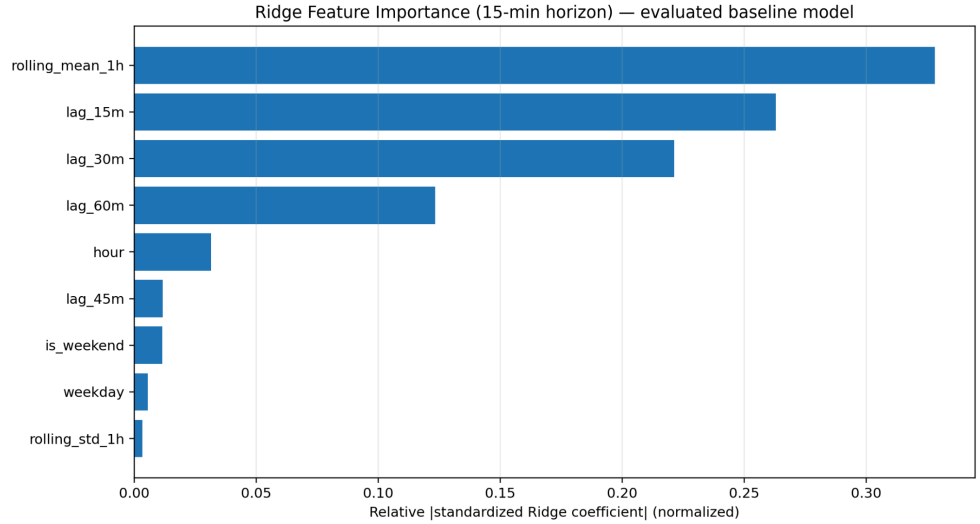
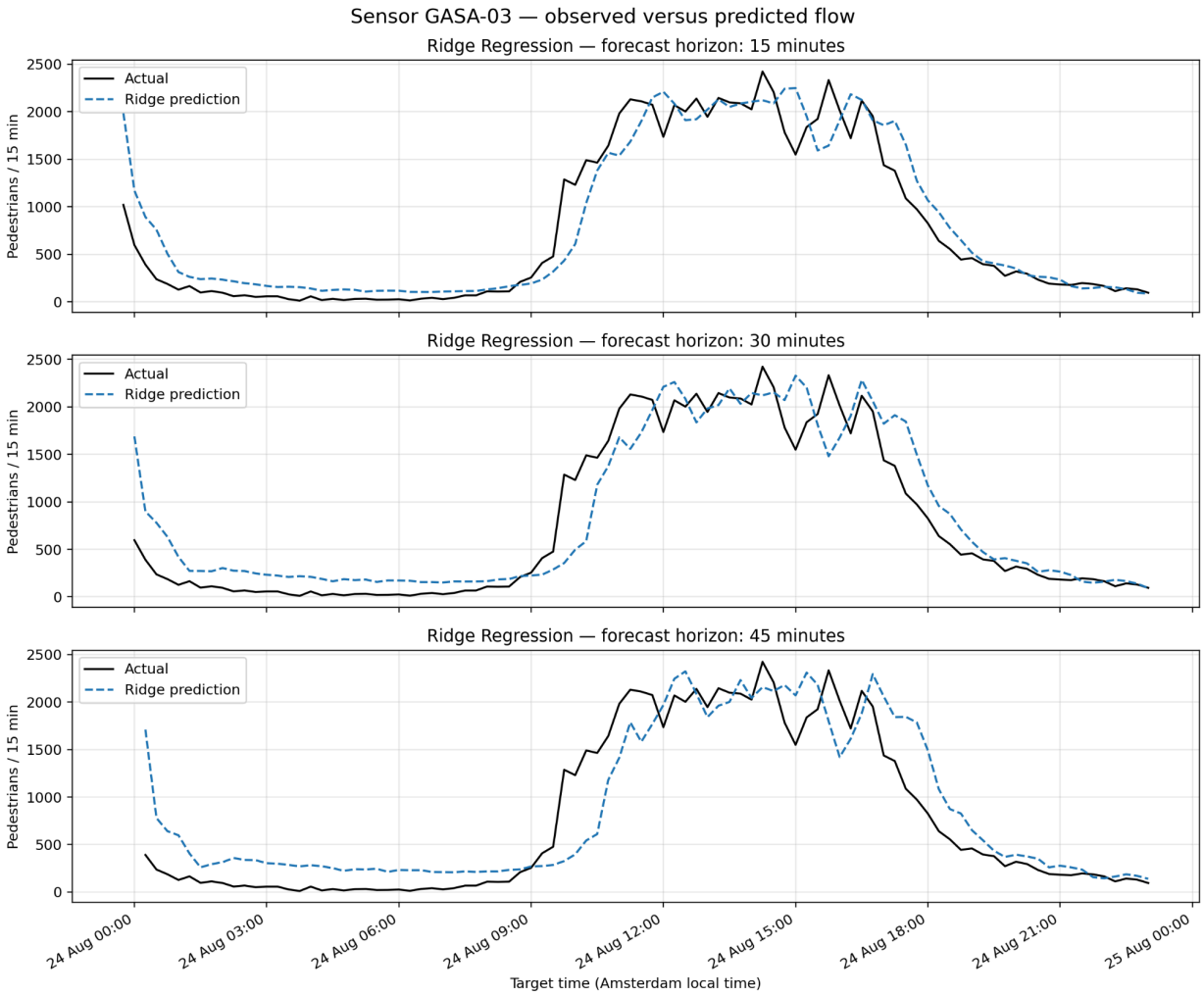


Figure 5.2: Ridge Regression evaluated baseline



Figures 5.3, 5.4, 5.5: Observed vs predicted flow - 15, 30, 45 minute horizon



5.2 XGBoost visualisation

In this section, we evaluate short-term crowd prediction models for the primary walkway corridor (sensor GASA-03, Oosterdoksbrug). The research objective is to assess whether incorporating temporal dynamics and external contextual information improves pedestrian flow forecasts across three distinct horizons: $t + 15$, $t + 30$, and $t + 45$ minutes.

We compare a linear autoregressive baseline (Ridge Regression) against a gradient boosting decision tree architecture (XGBoost Regressor). Both models are evaluated using standard quantitative error metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Subsequently, we examine feature importance to understand the relative contribution of historical pedestrian flows versus external contextual features.

Data preparation and temporal alignment

The raw crowd sensor observations are recorded at 3-minute intervals. In line with operational decision-making windows, the sensor counts are aggregated to uniform 15-minute intervals. Concurrently, weather observations (temperature, relative humidity, and rainfall) are resampled to the identical 15-minute grid using time-based linear interpolation.

Feature Engineering: Lags, Rolling Dynamics, and Targets

Supervised learning for time-series forecasting requires explicit construction of past observations as feature columns:

- Autoregressive Lags: Lag features (lag_15m through lag_60m) represent recent crowd volume at 15-minute steps prior to time t.
- Moving Window Statistics: The 1-hour rolling mean (rolling_mean_1h) and rolling standard deviation (rolling_std_1h) capture immediate trend trajectories and volatility.
- Calendar Embeddings: Categorical features (hour, weekday, is_weekend) account for circadian crowd rhythms.
- Forecast Targets: Future flow counts shifted by -1, -2, and -3 steps serve as the ground truth targets for horizons $t + 15m$, $t + 30m$, and $t + 45m$.

Train-test

To prevent data leakage and preserve temporal causality, random shuffling must not be applied. An 80/20 chronological split is implemented:

- Training Set: August 20, 01:00 to August 23, 23:15 (covering the event buildup and early days).

- Test Set: August 23, 23:30 to August 24, 23:00 (capturing high-density weekend operations).

Model Training & Performance Comparison across Horizons

We train Ridge Regression and XGBoost Regressor models across the three forecasting horizons, alongside a naive Persistence baseline. For each horizon, prediction errors are quantified using MAE and RMSE.

Horizon		Ridge MAE	XGBoost MAE	Ridge RMSE	XGBoost RMSE	Verbetering MAE (%)	
0	t + 15m	229.04	235.49	299.77	340.72	-2.8	
1	t + 30m	272.44	279.57	342.14	393.64	-2.6	
2	t + 45m	344.08	368.21	411.50	521.54	-7.0	

Table 5.2: XGBoost values

Performance Analysis:

Table 5.2 summarizes the forecast error metrics on the test partition:

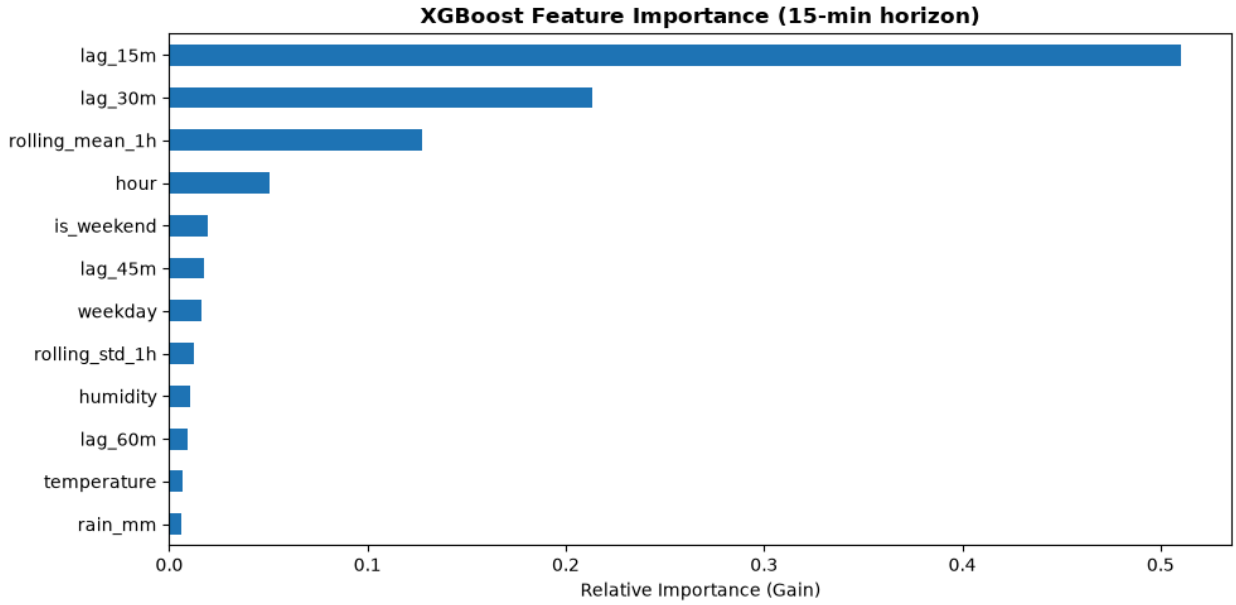
- Short-Term Accuracy ($t + 15m$): At the 15-minute horizon, Ridge Regression achieves an MAE of 229.04 (RMSE: 299.77), while XGBoost yields an MAE of 235.49 (RMSE: 340.72).
- Horizon Degradation ($t + 30m$ & $t + 45m$): As expected, predictive uncertainty increases with longer forecast horizons. By $t + 45m$, Ridge MAE rises to 344.08, and XGBoost MAE to 368.21.
- Ridge vs. XGBoost Dynamics: Ridge Regression performs marginally better across all three horizons (MAE difference of -2.6% to -7.0%). This occurs because pedestrian crowd series on high-capacity bridge corridors exhibit strong linear persistence during smooth transitions. Furthermore, tree-based models (such as

XGBoost) cannot extrapolate beyond the maximum values encountered in the training partition, resulting in slightly higher peak volume errors during unseen weekend spikes.

Feature Importance Analysis

To understand which variables drive the predictions, we evaluate the relative importance (Gain) of all features in the 15-minute XGBoost model.

Figure 5.6: XGBoost



Key Findings from Feature Importance:

- Dominance of Recent Lags: The immediate historical observation (`lag_15m`) and the rolling 1-hour average (`rolling_mean_1h`) constitute the vast majority of relative gain. This confirms that short-term crowd flows are overwhelmingly autoregressive; current congestion levels are the single strongest predictor of density 15 minutes ahead.
- Temporal Structuring: The time-of-day variable (`hour`) represents the next most impactful feature, establishing the baseline diurnal curve (morning rise, midday peak, evening decline).
- Contextual & Weather Variables: Meteorological variables (`temperature`, `humidity`, `rain_mm`) exhibit lower direct feature weights. Because major weather regime changes were concentrated in specific segments of the event, the model relies predominantly on flow momentum, with weather acting as a secondary moderator rather than a primary flow driver.

Added Value of External Context Information

The feature importance analysis shows that historical pedestrian flow measurements, particularly recent lag observations and rolling averages, play the largest role in predicting future pedestrian flows. In comparison, weather-related variables contribute less to the predictions. This suggests that recent crowd observations contain most of the information used by the models for short-term forecasting.

However, feature importance does not directly measure whether adding external information improves prediction accuracy. To determine this, the baseline and context-enriched models would need to be compared using identical test observations. Therefore, the current results suggest that historical pedestrian flow is the dominant predictor, but do not provide sufficient evidence to conclude that external contextual information improves or worsens forecasting accuracy.

6. Limitations and discussion

Discussion

In this window the tall ships are mostly static background rather than a changing signal, which limits what the crowd model can learn from the vessel features. The features describe how close ships lie to each sensor, so they can be added next to the weather variables to answer sub-question 1 (which context factors matter most) and sub-question 2 (added value over the baseline).

The vessel data shows that most ships lay still. Only one of the three sensors sees any variation in vessel features. At GASA-03 a ship is within 500 m in all 40 bins and within 250 m in 37, but never more than two ships at once. At GASA-04 only the same two ships come within 500 m and none within 250 m. At CMSA-GAWW-17 no ship comes within 1000 m so its features are constant and cannot explain any change in flow. No ship of 60 m or longer comes within 500 m of any sensor. A binary “big ship nearby” variable would therefore be zero for the whole window.

What this means for the research questions:

- The features hardly change within two hours, so the model cannot learn how the crowd reacts when a ship arrives or leaves. At most they act as a level per sensor, which the lag features of the crowd model already capture.
- In the XGBoost results of chapter 5.2 the lags (lag_15m, lag_30m) and the one hour rolling mean carry most of the gain, and the weather variables have low weights. We expect the vessel features to rank low as well, but this has to be tested, not assumed.
- The crowd models use 15-minute intervals, so the vessel features must be aggregated to the same grid. Two hours give eight intervals per sensor. With lags up to 60 minutes and horizons up to 45 minutes, that leaves too few samples to train on this window alone. The vessel features need to be evaluated at 3-minute resolution or combined with more days.

Design choices:

- Ship length instead of attractiveness. We use the summed length as a proxy for ship size. The vessel file has no attractiveness score, so none is used.
- Radii. The 250 m and 500 m radii were not tuned. GASA-03 and GASA-04 are 513 m apart, so their 500 m zones overlap and the same two ships count for both sensors.

Limitations

The vessel data cover only 20 tall ships for two hours on one day, so they give a thin basis for any conclusion about vessel activity and crowd flow.

- Two hours on one day. The window is Friday August 22, 16:00-18:00 local time: 40 bins per sensor and eight 15-minute intervals. The Sail-In parade, with its peak of tall ships on Wednesday August 20, falls outside it, so the features do not show the situation in which ships are most likely to draw a crowd.
- Only tall ships and sailing vessels. We selected only the SAIL fleet, so ferries, boat tours and other non-event traffic are not in the file, and neither are ships that attract visitors but do not belong to the fleet, such as cruise ships. The features therefore measure the presence of the SAIL fleet, not all vessel activity.
- Mostly moored ships. 67% of the pings are from ships lying still and only three ships were moving in most of their pings. There are hardly any arrivals or departures, so features based on movement or approach and the effect of a ship arriving cannot be estimated.
- Little spatial variation. CMSA-GAWW-17 has no ship within 1,000 m, GASA-04 has none within 250 m and GASA-03 has at most two ships within 500 m. The vessel effect can be estimated at one sensor only and on very few distinct values.
- Irregular reporting. The mean gap between pings of one ship is 90 s, the longest gap is 54 minutes and some ships report very rarely. A ship without a ping in a 3-minute bin is not counted in that bin, even if it is still lying at the same place, so the counts can be too low.
- Position of the ship. The reported position is one point, while ships are up to 115 m long. For large ships the hull lies tens of metres closer to a sensor than the distance suggests, and a ship that is 59 m long and 43 m away may in practice touch the quay next to the sensor.
- Straight-line distance. Haversine distance ignores canals, quays and walking routes. A ship 200 m away across the water can be further from the crowd than one 400 m away on the same quay.
- Correlation, not causation. A ship near a sensor and a high flow can both be caused by the time of day or the event itself. With features that hardly change, the model cannot separate the effect of ships from the daily pattern.

7. Conclusion

This research investigated whether external context factors, such as weather conditions and vessel activity, improve short-term pedestrian flow predictions during SAIL Amsterdam 2025. Ridge Regression and XGBoost were used to predict pedestrian flows 15, 30 and 45 minutes ahead, with performance evaluated using MAE and RMSE.

Firstly, we examined which factors contributed most to the predictions (SQ1). The results show that historical pedestrian flow measurements were the most important features in both models. Especially recent observations and rolling averages played a large role, while weather variables showed a smaller contribution. However, the additional predictive value of these external factors could not be fully established.

Secondly, we compared model performance across different prediction horizons (SQ2). Prediction errors generally increased for longer horizons. In Chapter 5.2, Ridge Regression performed better than XGBoost at all three horizons. At 15 minutes, Ridge Regression achieved an MAE of 229.04 compared to 235.49 for XGBoost. However, the separate Ridge analysis in Chapter 5.1 produced different results, making direct comparisons between all experiments difficult.

Furthermore, we investigated whether vessel activity could explain changes in pedestrian flows. We analysed 1,567 position reports from 20 vessels near three pedestrian sensors. Approximately 67% of the observations came from stationary ships. While vessels were detected near GASA-03 and GASA-04, none were observed within 1,000 metres of CMSA-GAWW-17. This shows that vessel activity had limited variation during the investigated two-hour period, making it difficult to determine its additional predictive value.

Finally, we examined model robustness (SQ3). Although a chronological train-test split was used, the limited observation period and incomplete weather data prevented a reliable evaluation under extreme conditions. Therefore, the findings cannot automatically be generalised to other events.

Bringing these findings together, historical pedestrian flow measurements appear to be the most useful predictors for short-term crowd forecasting during SAIL Amsterdam 2025. The additional predictive value of weather conditions and vessel activity could not be reliably demonstrated with the available data.

In conclusion, this study provides an initial basis for short-term pedestrian flow forecasting. Future research should include longer observation periods, more complete external datasets and additional sensor locations. Particularly, investigating major vessel movements such as the Sail-In parade could provide further insights into whether

external context information improves predictions and supports crowd management decisions.

8. Literature and AI statement

Artificial Intelligence (AI) was used as a supporting tool throughout this research project. AI, including Claude Code, assisted in writing and refining code, exploring relevant information, structuring research findings, and improving the clarity and quality of the report. Additionally, AI was used to enhance academic writing by reviewing grammar, spelling, sentence structure, and overall readability. All AI-generated suggestions, code implementations, and revisions were critically evaluated by the authors. The research decisions, analyses, interpretations, and final conclusions remain the responsibility of the research team.

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Contribution Statement

Be specific. Some of the tasks can be coding (expect everyone to do this), background research, conceptualisation, visualisation, data analysis, data modelling

Author 1: Dirk van Lent

Vessel data integration, writing the project report.

Author 2: Fabian de Grefte

GitHub setup, data cleaning, sensor selection, base dataset, data integration

Author 3: Daan Thijssen

Baseline linear ridge regression model and visualisation

Author 4: Hessel Molenaar

Introduction, Research Questions, Methodology, and general improvements.

Author 5: Ivo Weidema

XGBoost model and visualisation